

IFY STATISTICS | POPULATIONS, DATA TYPES AND SAMPLING

01 PEOPLE, GROUPS AND NUMBERS

Population	The whole group we want to study.
Sample	A part of that population selected for study.
Census	Collect data from every member of the population.
Parameter	A number describing the population ; often unknown.
Statistic	A number describing a sample ; may estimate a parameter.

Example: a fictional college

All 2,000 students = population.
200 surveyed students = sample.
120 of those 200 say yes: **60% is a statistic**.
The percentage of **all 2,000** saying yes is the population **parameter**.

Variable: a characteristic recorded about each person or item. **Data:** its recorded values.
Waiting time → 6.5, 4.2, 7.0 minutes.

Respondent: someone who answers the survey.
Selected and **responded** are different.

02 TYPES OF DATA

Categorical / qualitative	Labels or categories: programme, meal choice or student ID.
Quantitative / numerical	Counts or measurements: number of meals or waiting time.
Discrete	Quantitative values that are separate and countable : 0, 1, 2, ... meals.
Continuous	Quantitative values that can lie anywhere in a range; usually measurements : time or height.

Two common traps

A student ID is a **label**, even if written in digits.
Rounded height is still a **continuous measurement**.

03 FOUR SAMPLING METHODS

SIMPLE RANDOM

Every possible sample of the **chosen size** has the same chance. Randomly choose distinct IDs from a complete list.

Example: select 200 different IDs from 1–2,000. Friends and volunteers are not a simple random sample.

SYSTEMATIC

Choose a **random start**, then every **kth member**. The **interval k** is how far to move along the list.

Example: k = 10, start 7 → 7, 17, 27, ...
Check for repeating patterns in the list.

STRATIFIED

Divide into **strata** (one = **stratum**). Randomly select people from **every group**.

Proportional allocation: keep each group's population share. A 60% Business / 40% Engineering population gives **120 / 80** in a sample of 200.

CLUSTER

A **cluster** is a group, such as a class. Randomly select **some groups**; in one-stage sampling, include **everyone** in them.

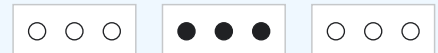
Example: choose 4 classes of 50 → 200 students. In **two-stage sampling**, sample people within the chosen clusters.

BOX = GROUP; FILLED DOT = SELECTED

Stratified



Cluster



Sampling frame: the list/source used for selection.

Representative sample: reflects relevant population features. Sample size alone cannot establish this.

IFY STATISTICS | FORMULAS, ERRORS, BIAS AND EVALUATION

04 FORMULAS AND EXAMPLES

N = population size; **n** = number selected;
k = sampling interval. Here $N = 2,000$; $n = 200$.

F1 PERCENTAGE OF RESPONDENTS

$$\frac{\text{Number giving the stated answer}}{\text{Number answering the question}} \times 100\%$$

120 of 200 respondents say yes:
 $120 \div 200 \times 100\% = 60\%$.
Use those answering **that question** as the whole.

F2 SAMPLING FRACTION

$$\frac{n}{N} = \frac{200}{2000} = 0.10 = 10\%$$

We select **10% of the population**. This is the share selected, not the share who answered yes.

F3 SYSTEMATIC INTERVAL

$$k = \frac{N}{n} = \frac{2000}{200} = 10$$

Random start from **1 to 10**. If it is **7**, add **10** each time:



First five IDs shown. Continue to **1,997** for 200 selections. These examples use whole-number intervals.

F4 PROPORTIONAL STRATIFIED SAMPLE

Number to select from one group =

$$\frac{\text{Group population}}{\text{Total population}} \times \text{total sample}$$

Business: $1,200 \div 2,000 \times 200 = 120$.

Engineering: $800 \div 2,000 \times 200 = 80$.

Check: $120 + 80 = 200$.

Then select students **randomly within each group**.
The formula gives **how many**; selection decides **who**.

Stratified does not always mean proportional.
Equal numbers from unequal groups can still form a stratified sample.

05 ERRORS AND BIAS

Sampling variation	Different random samples give different results by chance : 58%, 63%, 60%.
Sampling error	The sample-population difference caused by sampling . 63% versus a population value of 60% differs by 3 percentage points.
Bias	The process tends to push results away from the population value.
Selection / sampling bias	The selection process favours or excludes groups. Example: ask only library users about a college-wide policy.
Nonresponse bias	Respondents and nonrespondents differ in ways relevant to the study . Supporters may be more likely to reply.
Response bias	Wording or the survey situation encourages particular answers. Example: a leading question.
Measurement error	Recorded values differ from what they should represent. Example: an empty scale reads 20 g.

200 invited; 60 reply; 140 do not.
Missing replies alone do not prove bias.
No response is not an answer of "No".

06 EVALUATE AND IMPROVE

Accuracy	How close to the population value?
Precision	How similar across repeated samples?
Cost-effectiveness	Is the information worth the time and money?
Feasibility	Can we carry out the plan with our resources and access?

Improve: cover the full population; follow up nonrespondents; ask neutral questions; check equipment and procedures.

A larger, properly selected sample generally reduces **chance variation**. It does **not** fix excluded groups or biased wording.